EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Mid Term I Examination, Spring 2022

Course:	CSE 207- Data Structures, Section-1
Instructor:	Dr. Maheen Islam, Associate Professor, CSE Department
Full Marks:	40 (20 will be counted for final grading)
Time:	1 Hour and 30 Minutes

1. Given a linked list and two keys in it, write a function to swap nodes for two given k [Mark: 8]
Nodes should be swapped by changing links. Swapping data of nodes may be expe
in many situations when data contains many fields.

It may be assumed that all keys in linked list are distinct.

Example:

Input: 10->15->12->13->20->14, x = 12, y = 20

Output: 10->15->20->13->12->14

Input: 10->15->12->13->20->14, x = 10, y = 20

Output: 20->15->12->13->10->14

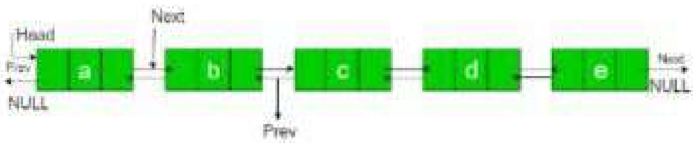
Input: 10->15->12->13->20->14, x = 12, y = 13

Output: 10->15->13->12->20->14

2. Suppose that, you are given a doubly linked list containing integer values. Write a fur [Mark: 8]
to rotate the linked list counter-clockwise by N nodes. Here N is a given positive i
and is smaller than the count of nodes in linked list.

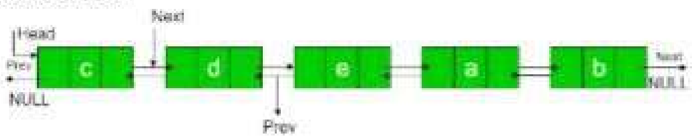
Example:

List:



N = 2

Rotated List:



3. Given an expression string exp , write a function to examine whether the pairs a [Mark: 8]
orders of "{","}","(",")","[","]" are correct in exp. For example, the program should pri
true for exp = "[()]{}{[()()]()}" and false for exp = "[(])"

4. Apply the algorithmic method to change the (X – A * (Y + Z) – P * Q) * R expressio [Mark: 8]
postfix expression using stack. Show each step of the conversion including stack
and postfix expression.

5. Write a program that dynamically allocates memory to store n number of intege [Mark: 8]
deletes the duplicate values from the list. For example, if the list contains - 10, 6, 2,
after deleting duplicate elements you will get the list - 10, 6, 2, 9. You have to w

function, called DeleteDuplicate (int n, int *p), where n is the total array size and
pointer variable that have array address to solve the problem.